

SocrateAI Scientific Agora Platform Engineering Report v14 Re-Run • FinOps • Epistemic Architecture

TuringAgent (Infrastructure)

SocratesAgent (Orchestration)

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Architecture Overview

The SocrateAI Agora v14 implements a 5-agent pipeline:

1. **SocratesAgent** — Dialectical orchestrator (PFC routing)
2. **TuringAgent** — GPU infrastructure (H100/A100/L4 Spot on GCP)
3. **GaloisAgent (SymBrain v11 Dual-Hemisphere)** — Mathematical conjecture generator (Left: symbolic PSLQ/LLL; Right: Gemini 2.5 Pro + Mistral MCTS)
4. **EulerAgent (Zero-Sorry Guillotine + Bayesian priors)** — Formal verifier
5. **PythagoreAgent (Probabilistic gap classifier)** — Lean 4 gap analysis

v14 Epistemic Upgrade: Zero-Sorry Guillotine

The v13 run contained a critical flaw: proofs marked **VERIFIED** contained 23–35 sorry stubs. This is mathematically impossible. In Lean 4, sorry is a kernel-level axiom; the compiler exits with code 0 even when present.

Fix Applied

1. Regex scan over Lean 4 code BEFORE compiler verdict is accepted

2. Any sorry occurrence: automatic INCOMPLETE, regardless of confidence
3. Orchestrator-level override: if Euler returns ≥ 0.85 but sorry count > 0 , confidence is capped at 0.70
4. Bayesian domain priors inform Euler's base confidence

Impact

29 v14 results had the guillotine applied.

FinOps Report

Metric	Value
Budget cap	\$400.00
v13 spend	\$15.26
v14 spend	\$12.94
Total spend	\$28.20
Budget remaining	\$371.80
Avg cost/problem	\$0.38
Problems attempted	34
Elapsed time	79.2 min

GPU Infrastructure

- **Priority:** H100 (us-central1) $\rightarrow$ A100 (us-central1-b) $\rightarrow$ 4 $\times$ L4 Spot
- **Instance type:** Spot/Preemptible (60–80% cost reduction)
- **Fallback regions:** us-east1, europe-west4, asia-east1
- **Service account key:** Auto-created and stored in Secret Manager
- **Auto-teardown:** TuringAgent scales to 0 after each problem
- **Scale-to-zero:** Enforced between problems (no idle GPU billing)

Bayesian Domain Priors (Euler v14)

These priors encode the difficulty of fully formalizing proofs in each domain:

Domain	Prior P(verified)
number_theory	5%
continuum_physics	10%
mathematical_physics	15%
discrete_geometry	20%
spectral_theory	20%
special_functions	25%
statistical_mechanics	30%
coding_theory	30%
combinatorics	35%

Conclusion

The SocrateAI Agora v14 demonstrates that LLM-driven mathematical conjecture generation, when combined with rigorous formal verification (Zero-Sorry Guillotine, Bayesian priors, Lean 4 gap analysis), can honestly characterize the frontier of AI formal mathematics. The system correctly distinguishes between:

- Problems where AI can generate plausible conjectures with formal sketches
- Problems at the genuine frontier of mathematical knowledge (Euler-Mascheroni, Feigenbaum constants) where even human mathematicians lack complete proofs